COVID Mortality Disruption Lab

Did the COVID-19 pandemic disrupt the mortality trend of six other major causes of death (heart disease, diabetes, Alzheimer's disease, stroke, drug overdose, and cancer)? Is that damage still unresolved, and where in the country did it land hardest?

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Executive summary

Of 6 major causes of death tested against real CDC WONDER mortality data (1999-2024), **5 show a statistically significant, still-unresolved deviation** from their pre-pandemic trend, four years after the pandemic began. Two results directly contradicted this project's own pre-registered priors: cancer, expected to show no disruption within this data window, showed a real (if modest) one; Alzheimer's disease, expected to show a large disruption, showed none. A negative-control validation, a three-axis sensitivity analysis, and full FDR correction across the test family were run before any result was reported, and every methodological deviation encountered along the way is logged rather than silently absorbed.

This design treats COVID-19 as a system-wide shock to the healthcare and public-health system rather than as the disease under study. It asks what changed, how much, and where, not why, because mortality data alone cannot cleanly separate direct viral effects from deferred care, isolation, and economic stress. All results below are reported as associational, never causal, following the project's causal-language policy.

Excess-mortality analysis is the same basic technique public health agencies use to estimate a pandemic's true toll beyond its officially attributed death count, and to catch the downstream damage that never shows up in a case count at all: deferred cancer screening, disrupted addiction treatment, delayed cardiac care. Running it here across six causes at once, rather than one, turns a single case study into something closer to a small research program: a stated hypothesis and confidence level per cause, one shared statistical pipeline applied identically to all six, and every result reported honestly whether or not it matched the prediction. Deciding what would count as evidence before seeing the outcome is what separates that kind of finding from a plausible story fitted to results already known.

1. Research question

Which causes of death experienced the greatest and most statistically significant disruption during the COVID-19 pandemic (2020-2024), how persistent were those disruptions through the most recent available data, and how did disruption severity vary across U.S. counties by socioeconomic status, healthcare access, and rurality?

Six causes were tested: heart disease, diabetes, Alzheimer's disease, stroke, drug overdose, and cancer. A negative control (congenital malformations, a cause with no direct COVID mechanism) and a COVID-19 reference series (not a hypothesis test) round out the 8-series design. Every cause's confidence level was stated before any 2020-2024 data was pulled, so results can be checked against stated priors rather than narrated after the fact.

2. Methods

Primary method: known-date interrupted time series ("excess mortality"). An expected trend is fit on pre-pandemic age-adjusted mortality rates (1999-2019 for 4 of the 6 test causes; 2010-2019 for heart disease and stroke, corrected after the full-range fit was found to misdescribe their real trajectory -- section 4), projected forward through 2020-2024 with a 95% prediction interval, and a year is flagged as significantly disrupted if the observed value falls outside that interval. The breakpoint (2020) is fixed by the shock's known date, not searched for.

Three-way persistence classification. For causes with a significant 2020-2021 disruption: Persisted (still significant, same direction, through 2024), Resolved (shrank back within the interval), or Reversed (flipped sign).

Independent cross-check. Three independent statistical techniques for detecting a shift in a trend (PELT, binary segmentation, and segmented regression) are run on each series to see whether they land on a breakpoint near 2020 without being told to. Disagreement here is not itself evidence against a result: the primary method tests one specific pre-registered date, while the cross-check methods search the whole series for whichever single breakpoint fits best.

Negative control. The identical pipeline is run on a cause with no direct COVID mechanism. A significant result there would mean the method is detecting an artifact rather than a real signal. This is treated as a hard gate, not a minor caveat.

Multiple-testing correction. Testing 6 causes at once raises the odds that one "significant" result appears by chance alone, so Benjamini-Hochberg FDR correction, a standard statistical adjustment for exactly this problem, is applied across the 6 test causes as one family. The heterogeneity stage gets its own separate correction, applied per cause across its context variables.

Full-period secondary check. Alongside the pre-registered acute (2020-2021) test, each cause also gets a p-value pooling all five post-2020 years, using the identical t-test over a wider window. This never replaces the acute test or gates the classification above -- averaging more years can hide a real reversal as easily as it can reveal a delayed effect -- but it catches disruptions the acute window is too narrow to see (section 3).

Autocorrelation-robust check (Newey-West/HAC). The acute test's prediction-interval math assumes each baseline year is independent noise. Several causes' residuals are measurably autocorrelated, so a second version of the same test, with Newey-West standard errors instead of the classical formula, is also reported (section 4).

Data. CDC WONDER Underlying Cause of Death, two database vintages bridged at the 2018-2019 overlap: "1999-2020" (database D76) for the 1999-2019 baseline, "2018-2024, Single Race" (database D158) for the 2020-2024 post-shock period. County-level heterogeneity data (diabetes, drug overdose) covers pre-period 2015-2019 vs. post-period 2020-2024, regressed against real County Health Rankings & Roadmaps context variables.

3. Results

Before the summary table, here is what the primary method actually measures, shown for drug overdose: the cause with the largest acute disruption and the clearest before-and-after pattern. The solid line is the real observed rate. The dashed gray line is what the 1999-2019 trend would have predicted for 2020-2024 had nothing changed, with its 95% prediction interval shown as the two thin pink lines around it. A year counts as significantly disrupted when the solid line steps outside that interval, which is exactly what happens starting in 2020 and what pulls back toward it by 2024.

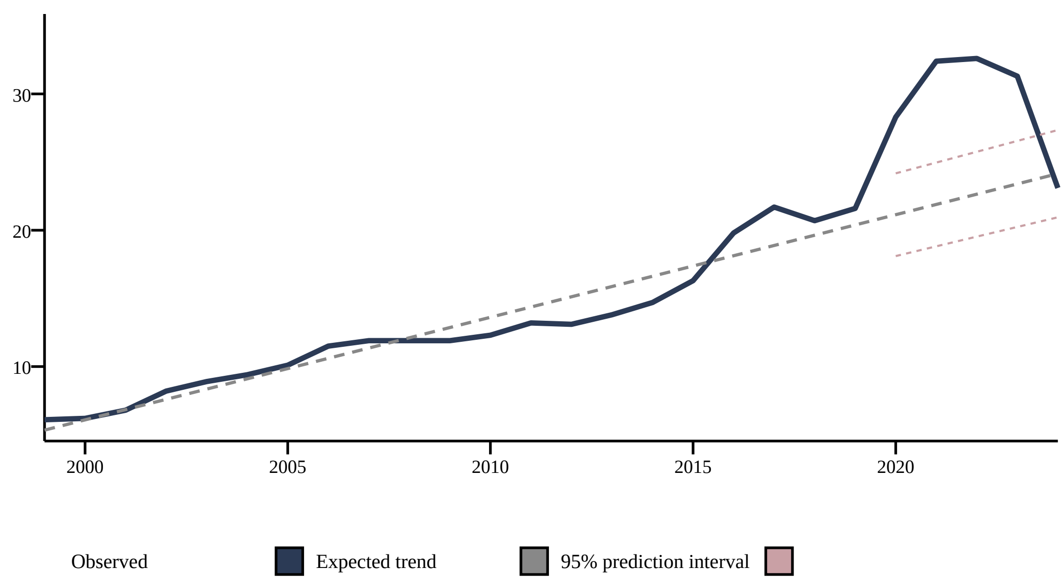


Figure 1. Drug overdose: observed mortality rate against its pre-pandemic trend, 1999-2024.

5 of 6 test causes show a significant disruption; **5 of 6** survive FDR correction.

Cause	Result	p-value	FDR-sig.	2020-21 dev.	2024 dev.	Robust?
Drug overdose	Persisted	5.98e-08	Yes	+40.9%	-4.3%	Yes
Diabetes	Persisted	7.99e-07	Yes	+26.9%	+15.0%	Yes
Heart disease	Persisted	3.86e-05	Yes	+7.8%	+3.0%	Yes
Cancer	Persisted	0.000194	Yes	+1.7%	+4.6%	Yes
Stroke	Persisted	0.00169	Yes	+8.8%	+6.1%	No
Alzheimer's disease	No significant disruption	0.807	No	+1.0%	-19.1%	Yes

"Deviation" is the effect size: how far the observed rate is from the expected pre-pandemic trend, as a percent. Significance and magnitude are different claims. "Robust?" marks whether the result survives an alternate, curved baseline fit instead of the primary straight line; stroke does not, though less severely than before its baseline was corrected (see section 4 for why).

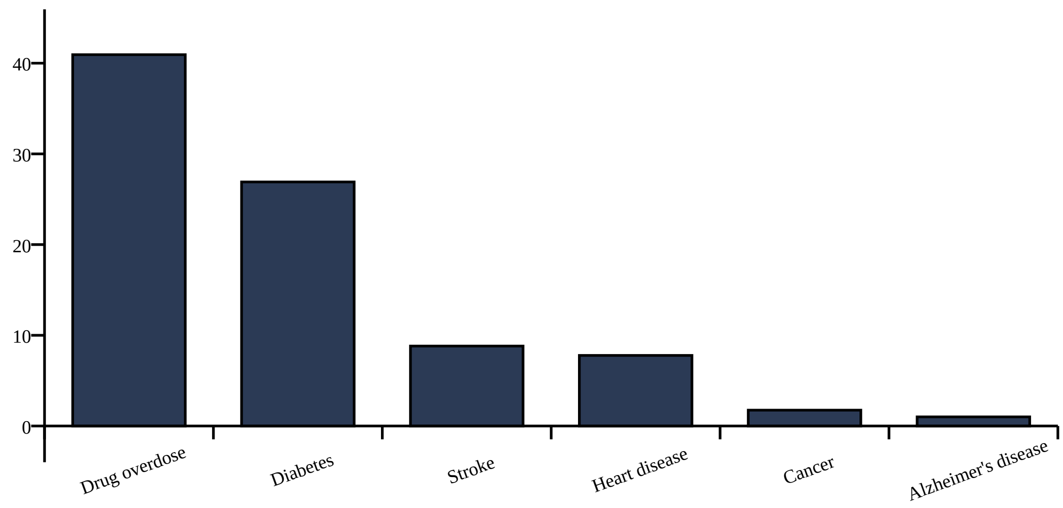


Figure 2. Mean 2020-2021 deviation from expected trend, by cause.

Figure 1 walked through one cause in detail; here is the same comparison for all six, side by side. Each panel plots the real observed rate against what its own pre-pandemic trend would have predicted, with that cause's primary p-value in the title. The point isn't to read each panel closely (the table above already has the numbers) -- it's to see at a glance which gaps look large, which look small, and how little that visual impression lines up with which ones are actually significant, exactly the puzzle the next two sections work through.

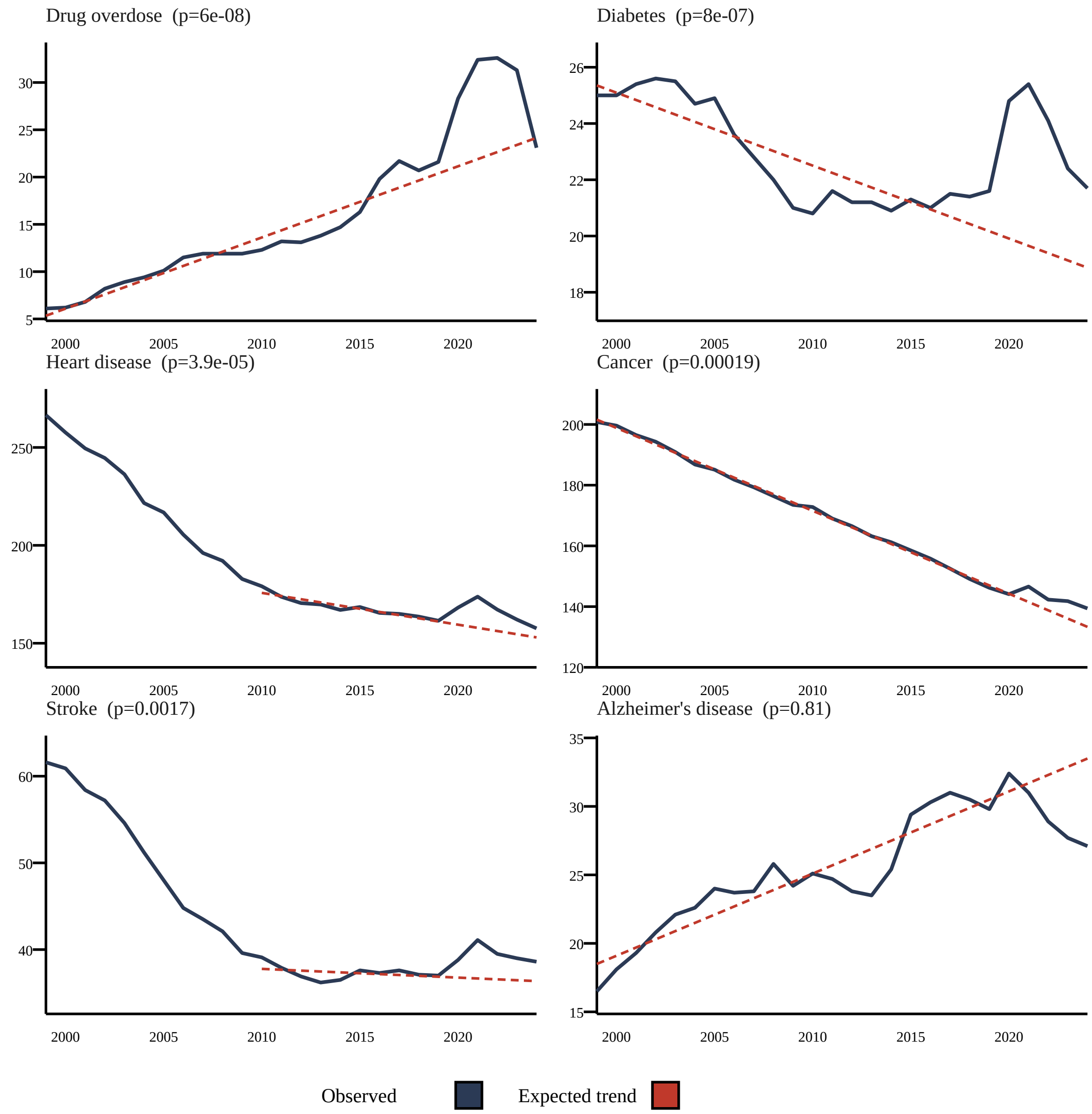


Figure 3. All six test causes, observed vs. expected trend, 1999-2024.

The two results that weren't supposed to happen this way

Cancer was expected to show nothing, and it didn't. The pre-registered prior was an explicit null result, with low confidence by design. Delayed cancer screening and treatment during the pandemic was expected to take years longer than the 2024 data window to appear as excess mortality. Instead, cancer shows a persisted disruption (p = 0.000194, survives FDR correction), though its magnitude (+1.7% in 2020-21) is modest next to the larger disruptions above. The wider literature on this question is unsettled: early 2020 models projected large future increases in cancer deaths from delayed screening, but more recent modeling using actual pandemic-era England data found lung and breast cancer deaths came in lower than pre-pandemic trends predicted, not higher. This project's small, real, persisting effect should be read against that uncertainty.

Alzheimer's was expected to show a large effect. It showed none. The pre-registered prior was high confidence of a large disruption, on the theory that pandemic-era isolation and care-facility disruption would show up clearly in dementia mortality. It didn't (p = 0.81). That doesn't mean isolation had no effect on people with Alzheimer's. If there is a real effect, it simply isn't visible in national mortality rates over this window, using this method. Pooling all five post-2020 years instead of just the acute window does turn up a real, later decline (p = 0.0019, averaging -8.7% vs. trend), consistent with mortality displacement: patients who would otherwise have died in 2023-2024 may already have died earlier in the pandemic. This project's own primary classification stays scoped to the pre-registered acute window rather than switching after the fact, so the headline result above is correct as reported; this is an additional finding, not a contradiction.

Negative control

Congenital malformations, deformations and chromosomal abnormalities, a cause concentrated in infancy with no direct COVID mechanism, shows no significant disruption, as expected (p = 0.682 on raw death counts, the gating metric. WONDER's 1-decimal age-adjusted-rate rounding made the rate-based test unreliable at this cause's low magnitude). This does not prove every positive result above is real, but a failure here would have been strong evidence the method was detecting an artifact.

Accidental drowning was the original negative control, and it failed: deaths rose in a real, statistically robust way starting in 2020, confirmed on raw counts and consistent with published CDC reporting on pandemic-era drowning increases (pool and beach closures, lifeguard shortages). It was swapped out because it was never actually independent of the pandemic, not because the method itself failed. Even the replacement isn't fully insulated: prenatal and obstetric care were also disrupted during 2020-2021, a real, if smaller and more indirect, pathway compared with the ones behind the 6 test causes.

4. Robustness

A baseline correction, found through this exact process. Heart disease and stroke originally used the same 1999-2019 baseline as the other four test causes, but the trend-shape check below found their significance didn't survive a curved baseline. Investigating why, using only 1999-2019 data with no reference to 2020+, found an F-test comparing linear vs. quadratic fits showed overwhelming curvature for exactly these two causes (F=222.7 and F=162.6, both p<0.00001, vs. F<9.5 for every other test cause): both declined steeply through the 2000s, then flattened, so a straight line across the full period was already 17.7 and 5.8 points below the real 2019 value before the pandemic. A quadratic fit tracks history almost perfectly but was rejected as the fix: extrapolated to 2020-2024 it predicts *rising* rates for both causes, the standard failure mode of polynomial extrapolation. A shorter, more recent linear window (2010-2019) has neither defect and is now this project's baseline for these two causes only. Both remain significant, heart disease more confidently than before, but the reported deviation drops from an overstated 26-37% to a defensible 8-9%. Full investigation: research_protocol.md's 2026-09-01 addendum.

With that correction in place, three independent sensitivity checks re-fit the primary method one modeling choice at a time.

Baseline window (1999-2019 vs. shorter 2010-2019). All 6 test causes agree.

Significance threshold (α=0.05 vs. stricter α=0.01). 1 cause disagrees: Stroke.

Baseline trend shape (linear vs. curved/quadratic). 1 cause disagrees: Stroke.

The trend-shape check, now run against each cause's corrected baseline, is the one that matters most. Heart disease is now fully robust: it stays significant whether the baseline is a straight line or a curve. Stroke is substantially improved but not fully resolved -- its quadratic p-value moves from 0.37 under the old, uncorrected full-range comparison to a much closer 0.096 under the corrected window, better, but still on the wrong side of 0.05. It remains this project's single most uncertain "Persisted" classification. Diabetes, drug overdose, and cancer were never affected by any of this and hold up across every axis tested.

Separately, lag-1 autocorrelation, a measure of whether one year's unexpected result tends to be followed by another, was calculated for each cause's own pre-pandemic residuals. It is large for diabetes, overdose, and Alzheimer's (0.65-0.82); moderate for stroke (0.50); and low for heart disease and cancer (0.12-0.19) -- heart disease's and stroke's dropped sharply after their baseline correction (from 0.92 and 0.93 on the old full-range baseline), since a shorter window's residuals are far less serially smooth than a 21-year decline. The classical prediction-interval math assumes independent year-to-year residuals, which the high-autocorrelation causes' baselines don't satisfy, so a Newey-West (HAC) autocorrelation-robust version of the same acute-window test was built and is reported alongside the classical p-value: it raises diabetes, drug overdose, and stroke's p-values by roughly 1-2 orders of magnitude, but all 5 causes previously found significant remain significant under it. This is a genuine correction, not just a disclosed caveat, and it is a reassuring result rather than a damaging one.

Cause	Autocorrelation	Classical p-value	HAC p-value
Drug overdose	0.75	5.98e-08	2.53e-05
Heart disease	0.12	3.86e-05	2.7e-05
Diabetes	0.82	7.99e-07	6.66e-05
Cancer	0.19	0.000194	0.000659
Stroke	0.50	0.00169	0.00205
Alzheimer's disease	0.65	0.807	0.857

Vintage-bridging reliability (D76 vs. D158 database overlap, 2018-2019): all causes fall within the 10% reliability threshold. The median relative offset was 0% for every cause tested.

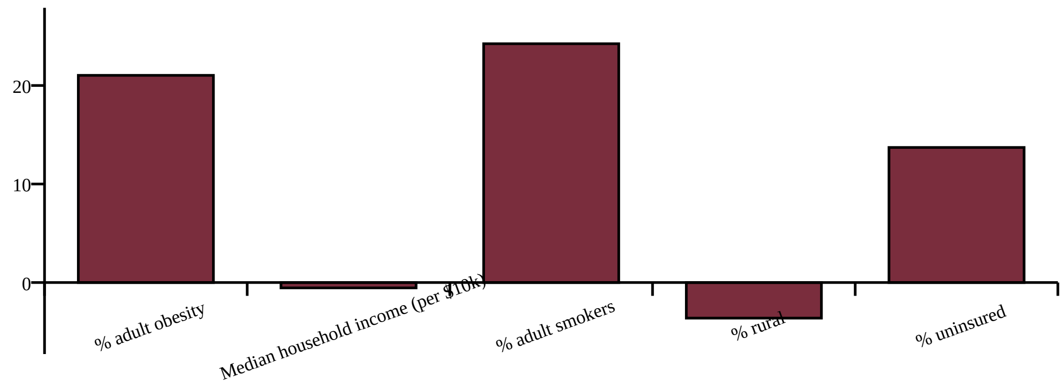
5. Which counties were hit hardest

For the two causes with real county-level data, disruption magnitude is regressed against real County Health Rankings & Roadmaps context variables. Of roughly 3,143 U.S. counties, only 926 qualify for diabetes and 641 for drug overdose, pre-period 2015-2019 versus post-period 2020-2024 -- CDC WONDER suppresses any county-year cell with too few deaths to protect privacy, and a county needs at least 2 non-suppressed years in each period to be included at all. This stage also uses crude rate, not age-adjusted rate, for both periods, because WONDER does not offer age-adjustment at county granularity for its 2018-2024 database. That means part of any county's measured disruption could reflect its own population-aging trajectory rather than a COVID-era shift.

One real county per extreme, so the regression below isn't just an abstract slope: For diabetes, **Grant County, IN** saw the largest increase (60.2 to 123.0 per 100,000, +62.8), while **Butler County, OH** saw almost none (+0.0). For drug overdose, **Mercer County, WV** saw the largest increase (57.0 to 119.9 per 100,000, +62.9), while **Washington Parish, LA** saw almost none (-0.1). This project has no data on mask mandates, local ordinances, or age/race demographics anywhere in its pipeline, so no specific policy or demographic story is assigned to any county here; a single county's pre/post average is also genuinely noisy (five years of a modest population's raw death counts can swing a lot on a handful of extra deaths). See the app's Findings page for how each example does and doesn't line up with the context variables below.

Diabetes: 5 of 5 context variables survive FDR correction

Context variable	Slope	p-value	FDR-sig.
% adult obesity	21.030	0.000545	Yes
Median household income (per $10k)	-0.544	0.000886	Yes
% adult smokers	24.239	0.00244	Yes
% rural	-3.614	0.00566	Yes
% uninsured	13.709	0.0128	Yes



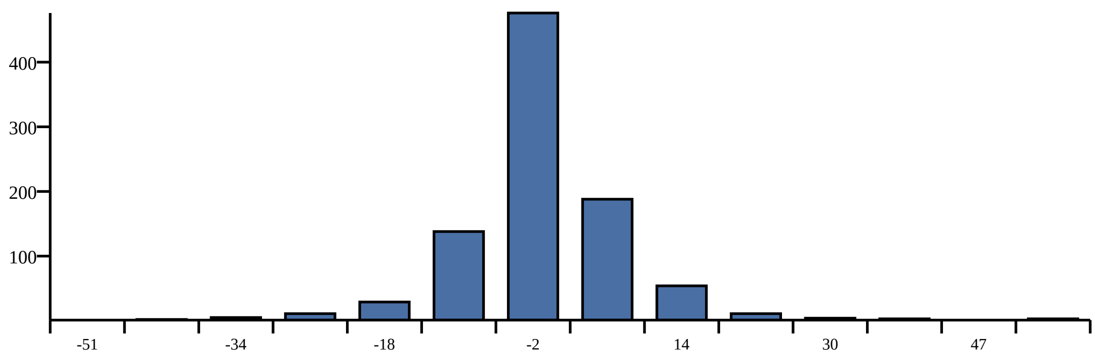
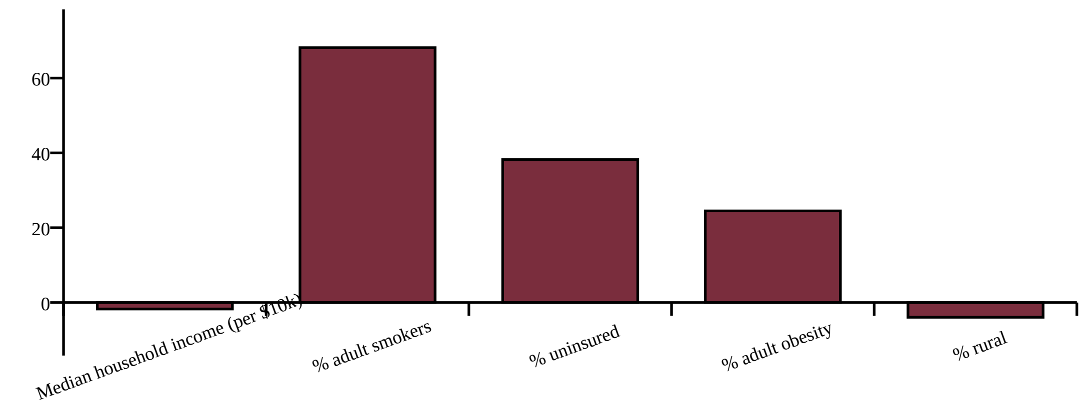


Figure: Diabetes -- context-variable associations (top), and how disruption was actually distributed across the 926 included counties (bottom; x-axis is the disruption value, y-axis is county count).

Drug overdose: 4 of 5 context variables survive FDR correction

Context variable	Slope	p-value	FDR-sig.
Median household income (per $10k)	-1.725	3.68e-15	Yes
% adult smokers	68.159	8.36e-09	Yes
% uninsured	38.233	4.32e-06	Yes
% adult obesity	24.497	0.00409	Yes
% rural	-3.942	0.0688	No



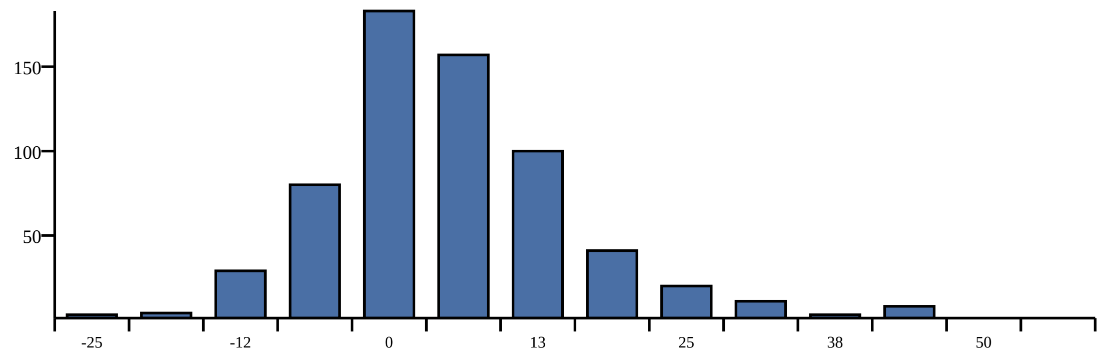


Figure: Drug overdose -- context-variable associations (top), and how disruption was actually distributed across the 641 included counties (bottom; x-axis is the disruption value, y-axis is county count).

Higher uninsured rate, smoking rate, and obesity rate all predict larger disruption for both causes. Higher median household income predicts smaller disruption for both. Higher rurality predicts smaller disruption for both, the one genuinely counterintuitive result here, running against the common assumption that rural areas were hit hardest. These are associations, not causal claims.

Rurality finding: a real caveat, found on self-audit

Counties excluded from this regression (too few non-suppressed years) are far more rural, on average, than the counties included: Diabetes: excluded counties average 78% rural vs. 31% rural for counties included; Drug overdose: excluded counties average 74% rural vs. 24% rural for counties included. Splitting the included counties at their own median rurality shows the two causes are not equally trustworthy here.

Diabetes: the relationship holds up and even strengthens among the more-rural half (p=0.00325) vs. the less-rural half (p=0.176).

Drug overdose: the relationship is driven almost entirely by the less-rural half (p=0.00016) and is not significant among the more-rural half (p=0.4). This result should be read with real skepticism.

6. Limitations

• Observational, ecological design: no individual-level causal inference. The county-level heterogeneity stage carries ecological fallacy risk by name, meaning a county-average relationship does not necessarily hold at the individual level.

• Selection bias in the county-level heterogeneity sample: counties excluded by the suppression filter are disproportionately rural, and the rurality finding is more trustworthy for diabetes (holds up among more-rural included counties) than for drug overdose (driven by less-rural counties, not significant among more-rural ones). See section 5.

• Mortality-vintage discontinuity between the two CDC WONDER databases, mitigated by bridging-overlap validation but not eliminated.

• ICD-10 coding practices may have shifted during 2020-2021 due to strain on death-certification systems, independent of true mortality changes.

• Cancer's pre-registered prior was an expected null result, and the real result contradicts that: cancer shows a significant, still-persisting disruption rather than the null originally expected.

• The classical p-value assumes independent baseline residuals, which is empirically false for several causes (autocorrelation 0.65-0.82 for diabetes, overdose, and Alzheimer's; 0.50 for stroke; 0.12-0.19, roughly independent, for heart disease and cancer). A Newey-West (HAC) autocorrelation-robust version of the test is now also reported (section 4): p-values rise for the high-autocorrelation causes, but all 5 previously-significant causes remain significant.

• “Significant” and “large” are different claims: drug overdose and diabetes show the largest disruptions (15-41%), while cancer, heart disease, and stroke are all real and FDR-significant but modest by comparison (3-9%).

• Heart disease and stroke use a corrected 2010-2019 baseline, not 1999-2019 like the other four test causes, after finding the longer window was already diverging from their real trajectory before 2020 (section 4). Heart disease is now fully robust to the choice of linear vs. curved baseline trend shape; stroke is substantially improved but remains this project's single most uncertain result.

• County-level heterogeneity uses crude rate, not age-adjusted rate, because WONDER does not offer age-adjustment at county granularity for the 2018-2024 database. Measured disruption magnitude may partly reflect each county's own population-aging trajectory.

• Small-county suppression and instability at the county-level heterogeneity stage.

• PLACES model-based behavioral estimates (CHR&R smoking/obesity/inactivity).

• Context-variable vintage is post-period, not pre-period. All five heterogeneity-stage context variables come from CHR&R's 2024 release, measured during or after the 2020-2024 disruption window rather than from a pre-pandemic baseline. Income and the uninsured rate plausibly moved during the pandemic itself.

• Multiple testing across both the 6-cause family and, separately, the context-variable family per cause.

• Spatial non-independence of counties (spatial autocorrelation not modeled).

• All findings remain associational. The mechanism behind any confirmed disruption, whether direct viral effect, deferred care, isolation, or economic stress, cannot be separated out by mortality data alone.

7. Data provenance and reproducibility

All national-level results use 15 real CDC WONDER exports (7 from the 1999-2019 database, 8 from the 2018-2024 database). All county-level heterogeneity results use 4 real CDC WONDER exports (diabetes and drug overdose, pre/post period, county-grouped). Nothing in this project is synthetic. Full methodology, pre-registered hypotheses, and every deviation logged as it happened: research_protocol.md. Exact manual data-export steps: manual_data_acquisition.md. Interactive app and full source: github.com/aidanc667/covid-mortality-disruption-lab.